



Microplastics abundance in marine sediments and stomach contents of benthic organisms

Alishon Yuliana Estrada-Stabilito · Karen Yicelth Morales-Duarte ·
Carlos David Rodríguez-Amaranto · Daniel Leonardo Ramírez-Orozco ·
Luis A. Gómez-Lemos

Received: 29 October 2025 / Revised: 25 January 2026 / Accepted: 5 February 2026 / Published online: 20 March 2026
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Abstract Microplastics are an emerging contaminant in marine ecosystems worldwide. This study evaluated the abundance of microplastics, their relationship with sediment type, and their ingestion by macroinvertebrates in the Eastern South Pacific Ocean and the Southern Caribbean Sea during the rainy and dry seasons. A granulometric analysis was performed applying a Pearson correlation matrix, and microplastics were extracted using the density flotation method and analyzed by type, color, and size. There was an average of 165 ± 22 microplastics/kg in the Eastern South Pacific Ocean and 296 ± 167 microplastics/kg in the Southern Caribbean Sea, however, there were no significant differences among regions or seasons. A negative relationship was found between microplastic abundance and coarser sediment, indicating that they retain fewer microplastics than finer sediments. Blue and red fibers smaller than 2 mm were

the most common microplastic types in both regions. There was a higher ingestion of microplastics in the Southern Caribbean Sea (22%) compared to the Eastern South Pacific Ocean (3%), and polychaetes were the main microplastic eaters. Our findings highlight the usefulness of marine polychaetes as bioindicators of microplastic pollution and suggest a relationship between the abundance of microplastics and the ingestion of microplastic by benthic organisms.

Keywords Marine pollution · Macroinvertebrates · Sedimentary bottoms · Eastern South Pacific Ocean · Southern Caribbean Sea

Introduction

Plastic particles smaller than 5 mm (microplastics) pose a threat to marine organisms and ecosystems worldwide (Browne et al., 2011). Their ability to widely disperse through ocean currents and accumulate in sediments makes them one of the most persistent pollutants with effects that can last for centuries (Thompson et al., 2004). The presence of microplastics has been reported in different marine environments including sedimentary bottoms (Knutsen et al., 2020). In sedimentary bottoms, the physical structure of the benthic community habitat is affected by the accumulation of microplastics (Blaskovic et al., 2017). In addition, microplastics act as vectors of pollutants such as heavy metals, polycyclic aromatic

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10750-026-06155-6>.

Guest editors: Massimiliano Scalici, Paula Sobral, Timothy Hoellein & Verónica Ferreira / Plastic Pollution in Aquatic Ecosystems.

A. Y. Estrada-Stabilito · K. Y. Morales-Duarte ·
C. D. Rodríguez-Amaranto · D. L. Ramírez-Orozco ·
L. A. Gómez-Lemos (✉)
Universidad Nacional de Colombia - Sede de La Paz -
Escuela de Pregrados, La Paz, Colombia
e-mail: lagomezle@unal.edu.co

hydrocarbons (PAHs), polychlorinated biphenyls (PCBs), organochlorine pesticides, nonylphenols, and phthalates (Vianello et al., 2013; Kesý et al., 2019). Microplastics can originate from the fragmentation of larger plastics (secondary) or be intentionally manufactured in microscopic sizes (primary), predominantly for use in cosmetic products and textiles (Hidalgo-Ruz et al., 2012).

The benthic macrofauna of sedimentary bottoms are mainly composed of annelids, mollusks, crustaceans, and echinoderms, play key ecological roles in nutrient cycles, and are part of the marine food chain (Aller and Cochran, 2019). Their high diversity and strong dependence on the substrate make these communities sensitive to environmental disturbances, and as such they are widely used as ecological indicators of environmental quality (Paredes et al., 2005; Zhang et al., 2023). The ingestion of microplastics by benthic organisms can cause behavioral changes, affect their ability to feed and reproduce (Hodgson, 2018), and facilitate the transfer of these particles to other trophic levels (Amelia et al., 2021).

Our study comprised two environmental and socioeconomic contrasting areas: an estuary in the Eastern South Pacific Ocean (ESPO) and the continental shelf in the Southern Caribbean Sea (SCS). Buenaventura Bay located in the ESPO and is a complex estuarine system fed by the Dagua and Anchicayá rivers (Cantera & Blanco, 2001). These rivers carry a great amount of fine sediment and organic matter to the bay (Lazarus & Cantera, 2007; Rincón et al., 2008). Buenaventura is the main port on the Colombian Pacific coast, with intense maritime and river traffic that increases the introduction of plastics into the sea (Vidal et al., 2021). People in this area mainly depend on small-scale fishing and aquaculture (Gamboa-García et al., 2020), therefore the presence of microplastics in sediments and benthic organisms can negatively affect food security and the well-being of these inhabitants. The Gulf of Salamanca located in the SCS has a narrow continental shelf where deep, cold, and nutrient-rich water arise, producing a seasonal upwelling (Castillo-Torres, 2002). In this area, the local atmospheric and oceanic circulation generates high hydrodynamics and these physical conditions, along with proximity to major urban and tourist centers, influence the dynamics of marine pollutants in the area (Galindo-Montero et al., 2023). Previous

studies indicate that coastal zones near urban settlements with intense port, fishing, and tourism activity have large amounts of microplastics in sediments (Galloway et al., 2017; Rangel-Buitrago et al., 2021). In summary, climatic and oceanographic patterns, along with local socioeconomic activities (such as tourism and fishing) may contribute to the accumulation of microplastics in the area (Ita-Nagy et al., 2022).

The aim of this study was to investigate the abundance and types of microplastics, their relationship with sediment features, and their ingestion by benthic macroinvertebrates in two contrasting coastal environments: an estuary in the ESPO and the continental shelf in the SCS. By comparing these regions, we seek to better understand the impact of microplastics in different oceanographic and socio-cultural contexts and contribute to the knowledge about their distribution, abundance, and effects on marine biota.

Material and methods

Study area

The study was carried out in two coastal areas of Colombia: The Gulf of Salamanca (SCS) and Buenaventura Bay (ESPO) (Fig. 1). The Gulf of Salamanca has a narrow, shallow continental shelf consisting of terrigenous sedimentary bottoms with lithoclastic and carbonate facies (Castillo-Torres, 2002), it is influenced by the intertropical convergence zone and the northeast trade winds (Restrepo & López, 2008). Climate varies throughout the year with a dry season from December to April, a rainy season from August to November, and a transition period from May to July (Ricaurte-Villota and Bastidas-Salamanca, 2017). Buenaventura Bay is a coastal estuary with depths ranging between 25 and 30 m and predominance of fine sediments such as clays and silts (Lazarus & Cantera, 2007). These sediments transported by the rivers that flow into the bay reach the seabed with a high content of organic matter (Gamboa-García et al., 2018). The rainy season occurs between September and October, the dry season between February and March, and the transition period in June and July (Thomas et al., 2014).

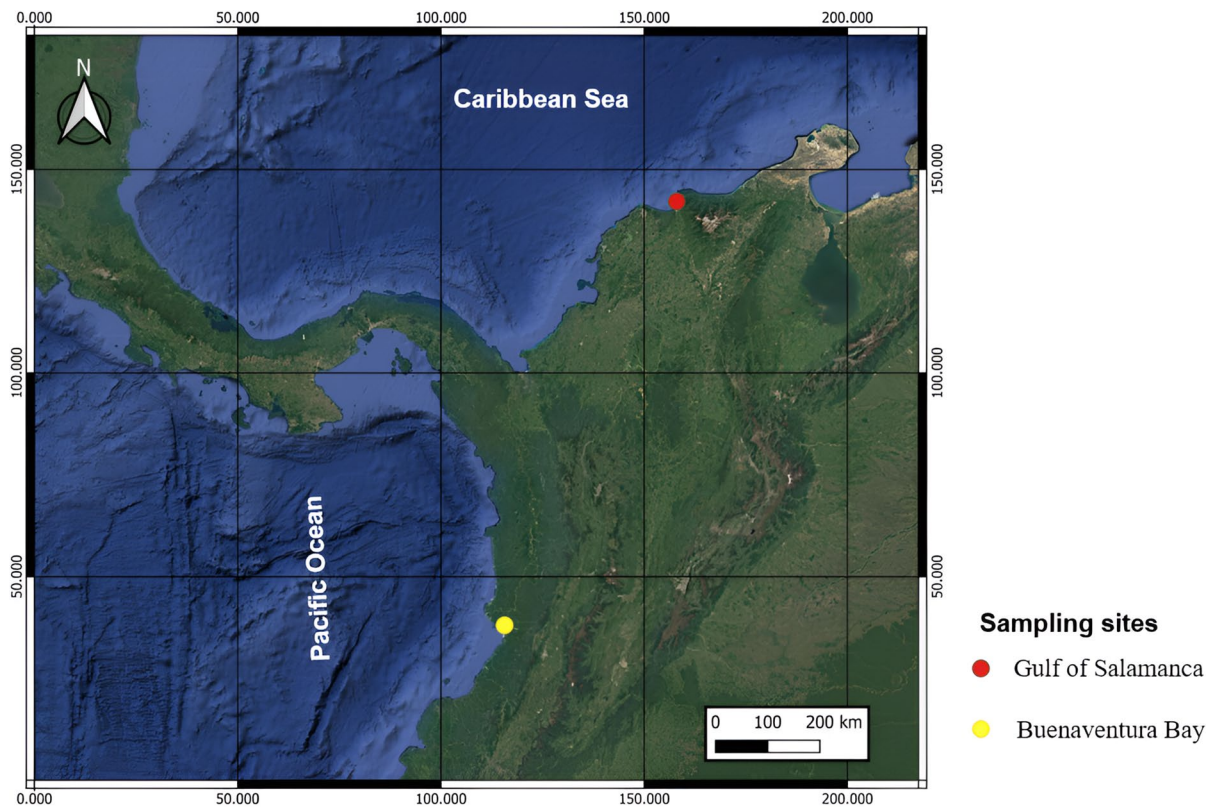


Fig. 1 Map of the study area with sampling sites in the Gulf of Salamanca (Southern Caribbean Sea) and Buenaventura Bay (Eastern South Pacific Ocean)

Field work

Sampling was carried out in both areas of study (SCS and ESPO) during two seasons: in the SCS during the rainy (October 2023) and transition season (May 2024), and in the ESPO during the rainy (November 2023) and dry season (March 2024). Two sampling sites were established at each area and three sediment samples (replicates) were taken from each site using a Van Veen grab with an area of 0.05 m². The coordinates of the sampling stations for each area are as follows: in the Gulf of Salamanca, sampling station 1 (11° 05' 51.4" N 74° 20' 10.5" W) and station 2 (11° 09' 24.8" N 74° 14' 05.7" W); in Buenaventura Bay, station 1 (3° 50' 29" N 77° 20' 55" W) and station 2 (3° 46' 9" N 77° 18' 6.8" W).

Sediment samples for granulometric and microplastic analysis were taken through the upper hatches of the Van Veen grab using a metal shovel. Samples were stored in aluminum containers to reduce the risk of external contamination and labeled with the sampling site, station, and replicate. A total of 24 sediment samples were taken (2 locations × 2 sites × 2 seasons × 3 replicates). Sediment samples for the study of benthic macroinvertebrates were stored in plastic bags and labeled with the sampling site, station, and replicate number. A solution of magnesium chloride (70 g/L) was added for 10 min to relax the organisms and facilitate their taxonomic identification in the laboratory (Baguley et al., 2006). Afterward, 10% formalin was added to preserve the organisms. We did not use the Rose Bengal dye to facilitate the separation of organisms from the sediment (Mason & Yevich, 1967), because it could modify the coloration of microplastics present in stomach contents.

Laboratory work

Granulometric analysis

Each sampling station had three replicates, with each consisting of 100 g of sediment oven-dried (Thermo Scientific-OMH100-S) at 60 °C for 72 h. The dried samples were then sifted using a sieve shaker (Filtration – FTS0200) with a set of six sieves (4000, 2000, 500, 250, 125, and 63 µm) for 10 min. After this period, the sediment retained in each sieve was weighed using a balance (Synergy-AS 220.X2 PLUS). The resulting values were used for the granulometric analysis (Blott & Pye, 2001).

Separation, extraction, and classification of microplastics

We followed the methodology proposed by Zhao et al. (2018) to separate and extract microplastics from sediment samples using the density flotation technique. The extraction process consisted of drying the sediments at 60 °C in an oven (Thermo Scientific-OMH100-S) for 72 h to eliminate moisture and achieve constant weight. Fifty grams (50 g) of dry sediment were mixed in 200 mL of saturated saline solution (0.24 NaCl g/mL) in 1000 mL glass Erlenmeyer flasks. The mixture was manually stirred with a glass mixer for 1 min and left to settle for 5 min. Later, the surface water containing the floating microplastics was extracted with a glass pipette. This process was repeated three times to maximize microplastic extraction.

Organic matter in the samples was eliminated by adding 1 mL of 30% hydrogen peroxide (H₂O₂) in 500 mL glass beakers and left to react for 24 h. Subsequently, 100 mL of the resulting solution was passed through a 1.0-µm pore-sized fiberglass filter (Whatman™) using a vacuum pump (Rocker 300). Later, each filter was covered with aluminum foil to prevent contamination with external microplastics and dried in an oven (Thermo Scientific OMH100-S) at 50 °C for 24 h. In addition, triplicate controls were made to quantify the abundance of microplastics in the laboratory. The procedure consisted of repeating the entire microplastic separation and extraction process, but without adding the sediment samples. Later, the microplastic abundance found in the controls was subtracted from the amount microplastic registered

in sediment samples from each region, sampling site, and climatic period. The result is the actual microplastic abundance measurement, eliminating microplastic contamination from the laboratory.

The dried filters were systematically examined using a stereo microscope (Carl Zeiss Stemi 305), following a zigzag pattern from left to right to achieve a complete inspection of the filter surface. We used four criteria for microplastic identification: (1) texture, evaluating whether the surface was smooth or rough which could indicate a different origin; (2) color, determining whether the hue was uniform or not; (3) presence or absence of cells or biological material, ruling out the possibility of organic remnants if cellular structures were observed; and (4) consistency, by gently pressing the material with tweezers to determine whether it was fragile or rigid (Hidalgo-Ruz et al., 2012). Microplastics were classified according to their shape into four categories: fibers, fragments, films, and pellets (Vásquez-Molano et al., 2021; Pasquier et al., 2022). It is important to mention that the method used did not allow us to identify the chemical composition of microplastics. Additionally, plastic particles were measured using a stereo microscope (Carl Zeiss Stemi 305) to verify that they did not exceed 5 mm in size.

Separation and taxonomic identification of benthic macroinvertebrates

Sediment samples were washed with freshwater and sieved through a 500 µm sieve. This procedure was performed to retain macroinvertebrates (invertebrates larger than 500 µm) and to eliminate smaller organisms and fine sediments (Holme & McIntyre, 1984). The retained material was distributed in small portions on a flat tray with water. Specimens were grouped by morphotype using a magnifying glass (Led Luxo) and stored in containers with ethanol (70%) for subsequent taxonomic identification.

Specimen identification was carried out down to the lowest possible taxonomic level using a stereo microscope (Carl Zeiss Stemi 305). For identification of organisms from the SCS, the following identification keys were used: Diaz & Puyana (1994) for mollusks, Benavides-Serrato et al. (2011) for echinoderms, Kensley & Schotte (1989) and Campos et al. (2005) for crustaceans, and Beesley et al. (2000) and De León González et al. (2011) for annelids.

For specimens collected in the ESPO, we used Keen (1971), Díaz & Puyana (1994), Daccarett & Bossio (2011), and García-Suárez et al. (2012) for mollusks; Brusca and Iverson (1985), Wetzer et al. (1997), and Campos et al. (2005) for crustaceans; García-Suárez et al. (2012) and Hickman (1998) for echinoderms; and Brusca and Iverson (1985), Laverde-Castillo (1986), Beesley et al. (2000), and LeCroy et al. (2000) for annelids.

Extraction of microplastics from macroinvertebrates stomach contents

We used the methodology proposed by Choy & Drazen (2013), dissecting the digestive tract to observe the gut contents in sterilized glass Petri dishes. Insulin-type needles (1 mm) were used for dissection and ethanol (70%) was added to prevent desiccation of specimens. For larger organisms, such as mollusks or echinoderms, a scalpel was used to open the digestive tract and access the stomach contents. To prevent microplastic contamination, the ethanol was pre-filtered using a vacuum pump (Rocker 300) with a 1.0- μ m pore-sized fiberglass filter (Whatman™) and subsequently stored in a 500 mL glass bottle. The extracted microplastics were counted, photographed, and data on the type, color, and size of the pieces were recorded (Vásquez-Molano et al., 2021; Pasquier et al., 2022).

Data analysis

Spatial and temporal variation in microplastic abundance in the sediments

To assess whether there were spatial and temporal differences in microplastic abundance within the study area, we performed a mixed ANOVA with interaction and nesting. This model allowed us to identify how microplastic abundance varies between geographic regions and seasons, taking into account the variability between seasons within each region. Regions (ESPO and SCS) and climatic season (rainy and dry/transition) were included as fixed effects. The fixed interaction consisted in evaluating the interaction between the factors region and climatic season to determine whether the effect of season varies between regions. Regarding the nested effect, the season factor is nested within each geographic region and is a

random effect but was included due to its influence on microplastic abundance. Initially, data did not present normality or homogeneity of variance, so they were logarithmically transformed. After logarithmic transformation, the data showed homogeneity of variance (Fligner–Killeen test, $P > 0.333$) and normal distribution (Shapiro–Wilk test, $P = 0.827$). These statistical procedures were performed using R-Studio software.

Relationship between microplastic abundance and sediment features

We analyze the sediment characteristics with the software Gradistat version 8 (Blott & Pye, 2001). To find linear relationships between parameters, a Pearson correlation matrix was constructed for both the SCS and the ESPO. The parameters included in the analysis were mean, sorting, skewness, kurtosis, and microplastic abundance. The mean denotes the average sediment grain size, which is influenced by the sediment source, the way of transportation, and the characteristics of the depositional environment (Sly et al. 1982). The mean is an important characteristic of marine sediments, giving insights into several environmental features, including geomorphologic physical processes occurring along the coastline (Visser, 1969). Sediment sorting indicates the degree to which sediment grain diameter varies relative to the mean particle size of a particular sample, and provides valuable information about the forces driving processes such as sediment erosion and accretion (Sahu, 1964). Sediment skewness measures the symmetry of grain size frequency distribution. Negative skewness indicates a dominance of coarser sediments, whereas positive skewness suggests a higher proportion of finer sediments (Duane, 1964). Consequently, negative skewness usually represents high-energy conditions that carry finer particles away, leaving coarser grains in place, while positive skewness denotes low-energy environments which favor the settlement of finer particles (Esmeralda & Patterson, 2025). Sediment kurtosis shows the sharpness of grain size distributions, highlighting the relationship between the classification of the extremes of the distribution and the central area, offering important insights about the characteristics of sediment distribution, whether it is leptokurtic (more peaked) or platykurtic (flatter) (Blott & Pye, 2001). The linear association between each pair of parameters was measured using Pearson's correlation

coefficient, using the `cor()` function in R version 4.3.1. The correlation matrix was constructed with the obtained coefficients; it is symmetrical and presents unitary values on the main diagonal. The values in the matrix are interpreted according to their magnitude and sign, identifying positive or negative linear correlations of variable intensity.

Results

Abundance of microplastics in sediments

The total abundance of microplastics was higher in marine sediments from the SCS compared to the ESPO (Table 1); however, the ANOVA test indicated that there were no significant differences in the average abundance of microplastics between geographic

regions, sampling stations, or climatic seasons (Table S1).

Characterization of microplastics

We found microplastics in sediment samples from both study regions. In the SCS, microplastics were mainly composed of fibers (89%), followed by fragments (5%) and films (4%). This pattern remained constant in both sampling campaigns and during both climatic seasons. The majority of particles were blue (36%) and red (25%), and the most common sizes were 1 mm (36%), < 1 mm (27%), and 2 mm (18%) (Fig. 2). Otherwise, in the ESPO, we observed an absolute predominance of fibers, representing 100% of the particles found. The most frequent colors were also blue (42%) and red (32%), and in terms of size, particles smaller than 1 mm

Table 1 Basic matrix of total microplastic abundance (microplastics/kg of sediment) in the study area, eliminating the effect of contamination in the laboratory

Climatic season	SCS		ESPO		Total season
	Station 1	Station 2	Station 1	Station 2	
Rainy	611	291	191	131	2804
	256	156	156	176	
	254	174	194	214	
Dry/transition	236	176	116	76	2096
	157	137	177	177	
	336	196	156	156	
Total station	1850	1130	990	930	4900
Total region	2980		1920		

SCS Southern Caribbean Sea, ESPO Eastern South Pacific Ocean

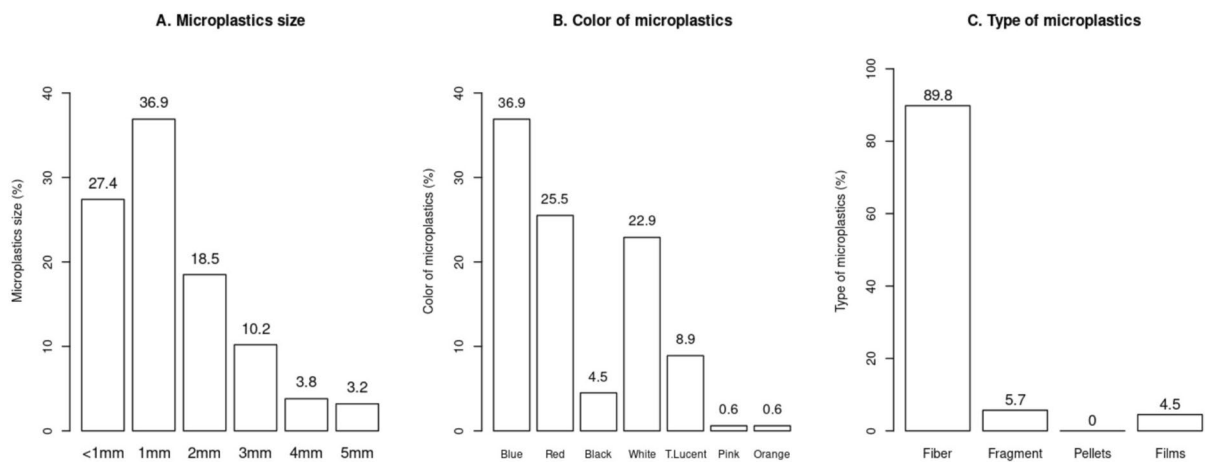


Fig. 2 Distribution histograms of **A** size, **B** color, and **C** type of microplastics found in marine sediments of the SCS

predominated (40%), followed by those of 2 mm (21%) and 1 mm (18%) (Fig. 3).

The results of the sediment analysis performed with the software Gradistat are summarized in Table S2. For the SCS, the mean grain size showed a negative correlation with the abundance of microplastics ($r = -0.475$), indicating that larger grain sizes are associated with lower amounts of microplastics. Sediment sorting showed a negative correlation with abundance ($r = -0.229$), suggesting that increasing sediment size reduces the amount of microplastics. Sediment Skewness and Kurtosis show the highest correlation in the matrix ($r = 0.953$); however, these variables are not related to microplastic abundance (Table S3).

For the ESPO, the mean grain size was also negatively correlated with microplastic abundance (although to a lesser extent than in the SCS, Table S4), indicating that larger grain sizes are also associated with lower microplastic abundances. However, contrasting the SCS, the mean grain size shows moderate negative correlations with skewness ($r = -0.522$) and kurtosis ($r = -0.488$), suggesting that as grain size increases, distributions tend to become less asymmetric and less leptokurtic (less concentrated around the mean). The relationship between skewness and kurtosis was very high ($r = 0.880$), confirming the close dependence between both parameters. However, both skewness and kurtosis showed weak correlations with microplastic abundance ($r = 0.211$ and $r = 0.073$, respectively).

Composition of the benthic macroinvertebrate assemblage

We collected a total of 231 specimens across the study area, 155 in the ESPO and 76 in the SCS (Fig. S1). The benthic macroinvertebrate assemblage was characterized by the dominance of the phylum Annelida in both study regions, with a lower presence of Mollusca and Crustacea. In the ESPO, the assemblage was dominated by polychaetes, with very few specimens of other phyla and an absence of Echinodermata (Fig. S2). In the SCS, although Annelida was also the dominant group, a greater relative taxonomic diversity was observed, with the presence of Mollusca and Crustacea, and Echinodermata only during the rainy season (Fig. S3).

Presence of microplastics in stomach contents of benthic macroinvertebrates

We found microplastics in the stomach contents of benthic macroinvertebrates in both study areas, although with variations in abundance, physical characteristics, and the taxonomic groups involved (Supplementary Information Figs. 4, 5). In the SCS, 22% of the specimens collected had microplastics in their digestive tracts, primarily annelids, which accounted for 70% of the observations. Within this group, families such as Nereididae, Eunicidae, Lumbrineridae, Paraonidae, Eulepethidae, and Goniadidae were identified. Microplastics were also detected in 23% of echinoderms, particularly in the families Amphiuridae

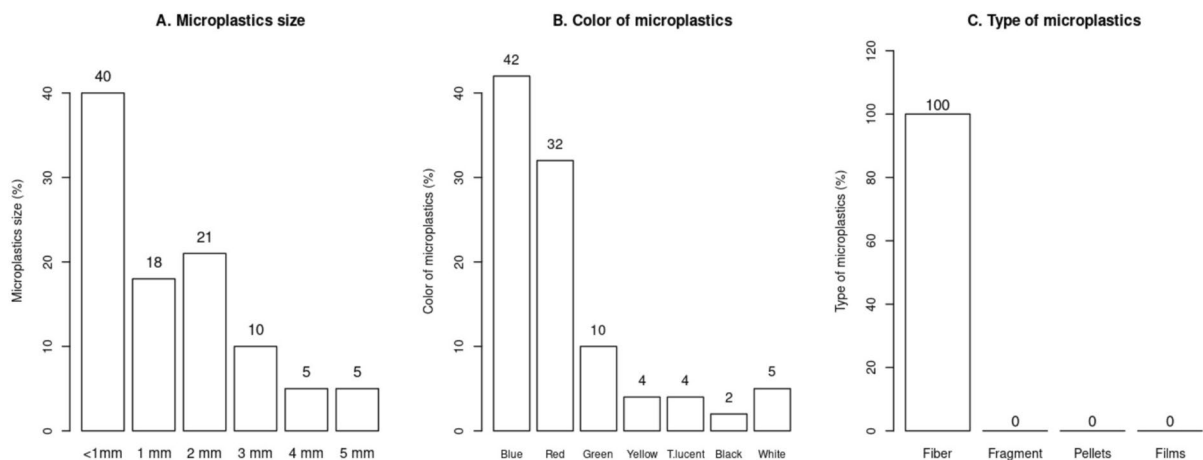
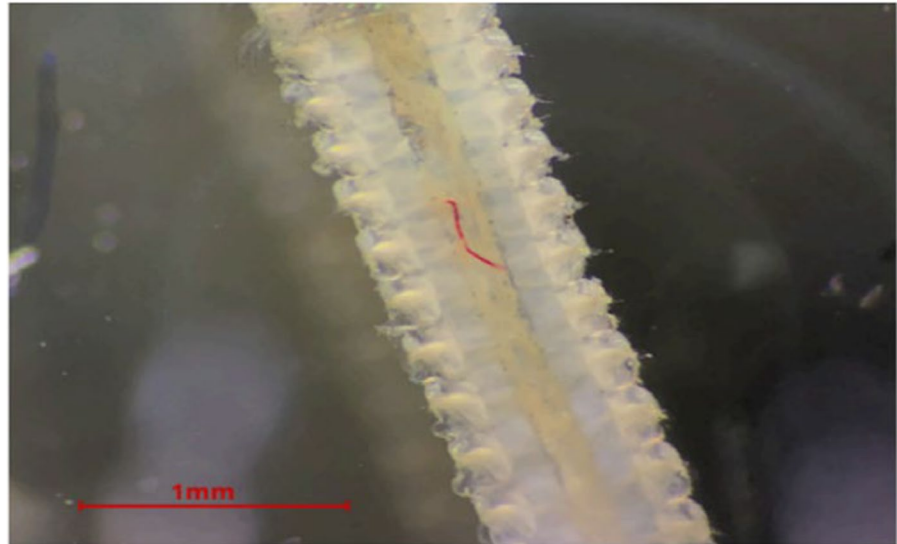


Fig. 3 Distribution histograms of **A** size, **B** color, and **C** type of microplastics found in marine sediments of the ESPO

Fig. 4 Microplastic (red fiber) found in the stomach contents of an annelid collected in the Gulf of Salamanca (SCS)



and Ophiomyxidae. Mollusks had a lower incidence (5%), limited to the family Veneridae. Regarding the type of microplastics, fibers predominated with 64%, while fragments accounted for 35% of the total particles. Regarding size, most of the recorded microplastics were less than 1 mm (68%), followed by 1 mm (22%) and 2 mm (8%) pieces (Fig. 4). We observed a lower microplastic ingestion in the ESPO, as only 3% of all specimens had microplastics in their stomachs. However, similarly than in the SCS, annelids were the most affected organisms, accounting for 66% of the observations. Microplastics recorded in the ESPO showed a wider size range, with 1 mm (40%), 2 mm (33%), and 3 mm (26%) being the most common.

Discussion

Abundance and distribution of microplastics in the sediments

Although we did not find significant differences in the average abundance of microplastics in sediments from the ESPO and the SCS, a higher total amount of microplastics was recorded in the SCS (Table 1). Previous Colombian studies have reported the same trend and attributed it to the greater industrialization, tourism, and fishing along the Caribbean coast of Colombia (Galindo-Montero et al., 2023). The differences in the incidence of microplastics between the SCS and the ESPO may be due to both anthropogenic

and natural factors. Inadequate waste management increases the pollutant load that reaches aquatic ecosystems (García et al., 2021). There are important differences in river discharges between the Gulf of Salamanca (SCS) and Buenaventura Bay (ESPO), influencing the distribution of microplastics in the marine environment. In the Gulf of Salamanca (SCS), the Manzanares and Gaira rivers show relatively low flows, averaging 1.8 and 2.7 m³/s, respectively (Invemar, 2008). Marine currents acting in the Gulf of Salamanca facilitate the transport of microplastics from freshwater sources. Northward and northeastward water flows penetrate the Gulf, creating small-scale cyclonic eddies that can contribute to microplastic accumulation (Franco-Herrera, 2005). On the other hand, Buenaventura Bay (ESPO) receives greater river input; with annual rainfall reaching 6980 mm, it is among the wettest places in the world (Cantera & Blanco, 2001). This high rainfall feeds the Dagua and Anchicayá rivers, with flows of 27 and 4.7 m³/s, respectively (ANLA, 2019), generating a conspicuous salinity gradient and a greater capacity to disperse pollutants such as microplastics in the estuarine environment (Molina et al., 2020).

The average abundance of microplastics in the ESPO (165 ± 22 microplastics/kg) was similar to that reported for the area by Vásquez-Molano et al (2021), who reported 194.9 ± 51.3 microplastics/kg for the year 2015. Our results represent an “intermediate” state of contamination compared with studies carried out in other countries. For example, lower

values have been recorded in Tuticorin on the south-east coast of India, with concentrations between 4.04 and 8.52 microplastics/kg (Esmeralda & Patterson, 2025). Concentrations closer to ours were recorded for the Southern Caspian Sea, from 25 to 330 microplastics/kg (Mehdinia et al., 2020) and on the Norwegian continental shelf, from 120 to 370 microplastics/kg (Knutsen et al., 2020). While cities with important solid waste management problems, such as Mumbai in India, show much higher values compared to our study, ranging from 4,900 to 14,500 microplastics/kg (Gurjar et al., 2022). Similarly, densely populated areas such as Tokyo (Japan) record value up to 21,100 microplastics/kg (Hashim et al., 2024). However, when making these comparisons, it should be noted that there may be methodological differences between studies in the phases of collection, separation, identification, and quantification of microplastics, which highlight the need to standardize methodologies. Additionally, it is important to mention that our study had methodological limitations as we did not chemically characterize the microplastics.

Relationship between sediment features and microplastic abundance

Previous studies indicate that sediment characteristics such as grain size, sorting, and texture influence the retention and distribution of microplastics in coastal environments (Hidalgo-Ruz et al., 2012; Mendes et al., 2021). In this study, we found a negative relationship between the mean grain size and the abundance of microplastics in both study regions. The mean represents the average sediment grain size, which is influenced by the sedimentary source, the mode of transportation, and the energy conditions in the depositional environment (Sly et al., 1982). Our results indicate that fewer microplastics accumulate in sedimentary bottoms where coarse grains predominate, compared to those dominated by fine particles. Sediment type and size are documented to have an important impact on microplastic abundance, with finer sediments typically showing the highest microplastic abundances (Vianello et al., 2013) compared to coarser sediments (Horton et al., 2017). This pattern has been reported in several marine ecosystems; for example, a recent study conducted in India to determine microplastic contamination in seagrass meadows found a moderate negative relationship

between mean grain size and microplastic abundance, suggesting that large sediment grains tend to correlate with low retention of microplastics (Esmeralda & Patterson, 2025). Furthermore, a study conducted in the supralittoral and infralittoral zones of Ireland found that polyethylene fibers and fragments were more abundant in fine-grained sediments than in coarse-grained sediments (Mendes et al., 2021). It has been suggested that microplastics are more efficiently retained by fine sediments, as they have a greater surface area relative to their volume (Enders et al., 2019). Furthermore, depositional environments where fine substrates predominate are characterized by low hydrodynamics, which favors the entrapment of microplastics in the sediment (Green & Johnson, 2020). Despite the pattern described by other authors and confirmed by our study, some investigations have not found any relationship between sediment type and microplastic abundance (Martins & Sobral, 2011; Alomar et al., 2016; Blaskovic et al., 2017; Renzi et al., 2018).

Microplastics features

Our results are consistent with previous studies conducted in similar marine environments. The dominance of fibers in the sediments of both regions reinforces the trend widely documented in the scientific literature (Song et al., 2014; Peng et al., 2018; Vázquez-Molano et al., 2021; Alves et al., 2022; Do et al., 2022; Zheng, 2022). For example, a study by Zhang et al. (2023) in Jiaozhou Bay (China) found that microplastics present in the sediments were predominantly black and blue fibers, with sizes smaller than 500 µm, results that coincide with those observed in our study.

Rangel-Buitrago et al. (2021) suggested that coastal areas adjacent to urban settlements where fishing activities occur tend to have higher concentrations of plastic fibers in their sediments. We observed this pattern in both study regions, since both Buenaventura Bay (ESPO) and the Gulf of Salamanca (SCS) are influenced by urban centers, intense fishing, and port activity (Invemar, 2008). In particular, in the case of the Gulf of Salamanca, its proximity to populated areas such as Santa Marta, where tourism, port, and fishing activities converge, suggests a relationship between anthropogenic sources and the presence of fibers in the sediment. In the study by

Woodall et al. (2014) in deep areas of the Atlantic Ocean, the Mediterranean Sea, and the Indian Ocean, fibers represented 89% of the identified microplastics, further consolidating the notion that this type of microplastic is the most common in marine environments. This dominance of fibers has been attributed to their detachment from synthetic textiles during washing processes, as well as the wear and tear of plastic materials used in fishing and domestic activities (Browne et al., 2011). In this study, in Buenaventura Bay (ESPO), microplastics in the sediments were observed to be exclusively fibers (100%), with sizes between 0.5 and 1.0 mm and predominantly blue and red. These results are consistent with those reported by Song et al. (2014), who also identified fibers as the predominant type of microplastic in marine sediments off Geoje Island, South Korea. This similarity reinforces the hypothesis that smaller particles tend to predominate in marine environments, as also suggested by Isobe et al. (2015, 2017) and Zhang et al. (2017).

In this study, blue, red, and green microplastics predominated in the ESPO, while blue, red, and white were more frequent in the SCS. The high abundance of blue microplastic has been widely documented in other studies and attributed to the wear and tear of textile waste and fishing nets; it has also been suggested that their high visibility in the sea may facilitate their ingestion by aquatic organisms (Vianello et al., 2013). On the other hand, the white color, although less abundant, is indicative of polyethylene, one of the most common polymers in plastic waste (Bharath et al., 2021). According to Choong et al. (2021) and Wicaksono et al. (2021), the yellowish hue may be related to photooxidation processes of plastics exposed to solar radiation, while transparent microplastics are usually associated with polypropylene, which is widely used for food packaging (Aslanzadeh & Kish, 2010). Additionally, the predominance of irregular pieces smaller than 2 mm in both study areas suggests that the majority correspond to secondary microplastics derived from the degradation of larger plastics (Corti et al., 2023). This fragmentation process occurs mainly due to the combined action of ultraviolet radiation and mechanical abrasion from waves (Thompson et al., 2004). The similarity in the physical features (shape and size) of the microplastics found in ESPO and SCS suggests common sources of contamination. These could be primarily related to

textile degradation, fishing, and port pollution, where polymers such as polyethylene (PE) are widely used (Zhang et al., 2023). This similarity in the types of microplastics found suggests that the production and consumption dynamics in the study area play an important role in the generation and dispersion of this pollutant in marine ecosystems.

Presence of microplastics in stomach contents of benthic macroinvertebrates

In this study, the presence of microplastics in the stomach contents of benthic macroinvertebrates was lower in the ESPO than in the SCS. In both regions, polychaetes showed a higher abundance of microplastics in their digestive tracts than other groups of macroinvertebrates. This pattern can be attributed to their primarily deposit-feeding habits and their close relationship with sediments, the place in which microplastics accumulate the most (Vecchi et al., 2021). Most deposit-feeding organisms feed by processing large amounts of sediment through their digestive system, ingesting a daily volume of sediment that can be equivalent to their own body weight (Lopez & Levinton, 2011; Gorska et al., 2019). Therefore, the concentration of microplastics in sediments directly influences the amount ingested by these organisms (Setälä et al., 2016; Hiki & Nakajima, 2022; Hashim et al., 2024). This pattern confirms the usefulness of deposit-eating polychaetes as bioindicators of microplastic contamination in marine sedimentary beds (Pagter et al., 2021).

Studies in highly urbanized areas such as Tokyo Bay (Japan) report microplastic ingestion by 90% of benthic macroinvertebrates (Hashim et al., 2024). Tokyo has been classified as the most urbanized and densely populated metropolitan area in the world (Demographia, 2023). In contrast, our results in Buenaventura Bay (ESPO) and the Gulf of Salamanca (SCS) indicate considerably lower microplastic ingestion by macroinvertebrates (3 and 22%, respectively). These notable differences may be associated with factors such as population density, the degree of urbanization (Narmatha-Sathish et al., 2020; Ben-Haddad et al., 2022), oceanographic conditions, and fishing activity (Mendes et al., 2021) among others. Tokyo, with a population close to 37 million people (Hashim et al., 2024), faces greater challenges in terms of environmental pollution than Buenaventura

(ESPO) and Santa Marta (SCS), whose populations are much smaller (324,643 and 566,650 inhabitants, respectively) (Alcaldía Distrital de Santa Marta, 2024; DANE, 2025). Lopes et al. (2023) conducted a study in Rio de Janeiro (Brazil) to determine the relationship between trophic ecology, urbanization, and microplastic ingestion, finding that 55% of benthic macroinvertebrates ingested microplastics. Additionally, the highest abundances of this pollutant were found in deposit-feeding species that inhabit areas with the highest levels of urbanization. However, it is important to mention that some studies have not found a clear relationship between the degree of coastal urbanization and the consumption of microplastics by benthic macroinvertebrates (Hodgson, 2018; Costa et al., 2021).

Regarding size, the majority of microplastics found in the digestive organs of benthic macrofauna in this study (both in the ESPO and SCS) corresponded to the smallest size categories (≤ 1 mm). Similar results were reported for the polychaetes *Sabella Spallanzanii* (Gmelin, 1791) and *Hermodice carunculata* (Pallas, 1766), which under natural conditions mainly ingest microplastics in a size range from 90 μ m to 1 mm (Vecchi et al., 2021). This pattern is consistent with that reported by other studies who have shown that the smaller the microplastics the greater the probability of being ingested (Botterell et al., 2019; Hashim et al., 2024). Microplastics can be mistaken for natural prey during feeding due to their similar sizes (Botterell et al., 2019). However, size selectivity has been observed in some invertebrates such as crustaceans (Desforges et al., 2015) and annelids (Hurley et al., 2017). Regarding the shape of microplastics, fibers were the most abundant particle type in stomach contents in both study regions (ESPO and SCS), a pattern commonly reported for different phyla (Davinson & Dudas, 2016; Carreras-Colom et al., 2018; Compa et al., 2018; Vecchi et al., 2021). This pattern is due to the fact that the most abundant type of microplastic in the ocean are fibers (Woodall et al., 2014; Peng et al., 2018; Do et al., 2022; Zheng, 2022).

Although our microplastic ingestion records are lower than those reported by other studies (i.e., Lopes et al., 2023; Hashim et al., 2024), it cannot be ignored that the consumption of this pollutant can generate multiple adverse effects in macroinvertebrates, such as alterations in behavior, feeding, development, and

reproduction (Hodgson, 2018; Amelia et al., 2021). Microplastics can also act as vectors of toxic substances such as heavy metals, antibiotics, and polycyclic aromatic hydrocarbons, increasing the risk of bioaccumulation and biomagnification throughout the marine food web (Kesy et al., 2019; Rusinque-Quintero et al., 2022). Additionally, they can also transport pathogens and invasive species (or their propagules) (Gallitelli et al., 2021).

Conclusions

The abundance of microplastics in sediments from the study area represents an intermediate pollution level compared with other places around the world. However, there are methodological differences between studies during collection, separation, identification, and quantification of microplastics, which highlight the need to standardize methodologies. We found a negative relationship between microplastic abundance and coarser sediment grains, indicating that they retain fewer microplastics than finer sediments. The most common microplastic types in our study were blue and red fibers smaller than 2 mm, this pattern can be due to the wear and tear of textile waste and fishing nets. The great majority of these fibers are secondary microplastics derived from the degradation of larger plastic pieces. Microplastic ingestion by macroinvertebrates was higher in SCS than in ESPO, and polychaetes were the main microplastic eaters in both regions. This result confirms the usefulness of polychaetes as bioindicators of microplastic contamination in marine sediments. Finally, we suggest a relationship between the abundance of microplastics in marine sediments, the degree of coastal urbanization, human population density, and the ingestion of microplastic by benthic macroinvertebrates.

Acknowledgements We wish to thank Professor Guillermo Duque Nivia and Econacua members for helping us with the field work. We thank Bonnie Lewis for providing insightful comments and edits to this manuscript. Additionally, we acknowledge the anonymous reviewers for their valuable corrections and suggestions to improve the manuscript. This study was funded by Universidad Nacional de Colombia, project number 57700.

Author contributions Alishon Yuliana Estrada-Stabilito, Karen Yicelth Morales-Duarte, and Luis A. Gómez-Lemos contributed to the study conception and design. Material

preparation, data collection, and analysis were performed by all authors. The first draft of the manuscript was written by Alishon Yuliana Estrada-Stabilito and Karen Yicelth Morales-Duarte and all authors commented on the previous versions of the manuscript. All authors read and approved the final manuscript.

Funding Open Access funding provided by Colombia Consortium. This study was funded by Universidad Nacional de Colombia—Project number: 57700 (Luis Alonso Gómez Lemos).

Data availability The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

Code availability Not applicable.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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References

- Alcaldía Distrital de Santa Marta, 2024. Plan de Desarrollo Distrital 2024–2027: Santa Marta. Retrieved from <https://www.santamarta.gov.co/sites/default/files/PDD-Santa-Marta-2024-2027.pdf>
- Aller, R. C. & J. K. Cochran, 2019. The critical role of bioturbation for particle dynamics, priming potential, and organic C remineralization in marine sediments: local and basin scales. *Frontiers in Earth Science* 7: 157. <https://doi.org/10.3389/feart.2019.00157>.
- Alomar, C., F. Estarellas & S. Deudero, 2016. Microplastics in the Mediterranean Sea: deposition in coastal shallow sediments, spatial variation and preferential grain size. *Marine Environmental Research* 115: 1–10.
- Alves, F., L. M. Pinheiro, C. C. Bueno, V. Agostini, L. Perez, E. H. L. Fernandes, J. Weschenfelder, A. Leonhardt, M. Domingues, G. L. L. Pinho & F. García-Rodríguez, 2022. The use of microplastics as a reliable chronological marker of the Anthropocene onset in Southeastern South America. *Social Science Research Network*. <https://doi.org/10.2139/ssrn.4209135>.
- Amelia, T. S., W. M. Khalik, M. C. Khalik, Y. Ong Shao, T. Pan & H. P. Bhubalan, 2021. Marine microplastics as vectors of major ocean pollutants and their hazards to the marine ecosystem and humans. *Progress in Earth and Planetary Science*. <https://doi.org/10.1186/s40645-020-00405-4>.
- Aslanzadeh, S. & M. H. Kish, 2010. Photo-oxidation of polypropylene fibers exposed to short wavelength UV radiations. *Fibers and Polymers* 11: 710–718. <https://doi.org/10.1007/S12221-010-0710-8>.
- Autoridad Nacional de Licencias Ambientales (ANLA). (2019) Resolución No. 01193, 25 de junio de 2019 – “Por medio de la cual se imponen unas medidas de manejo ambiental y se adoptan otras determinaciones”, basada en la visita técnica del 13 al 17 de mayo de 2019. Retrieved from <https://www.anla.gov.co/images/documentos/resoluciones/2024-05-30-anla-resolucion-01193-25062019.pdf>
- Baguley, J. G., P. A. Montagna, L. J. Hyde, R. D. Kalke & G. T. Rowe, 2006. Metazoan meiofauna abundance in relation to environmental variables in the northern Gulf of Mexico deep sea. *Deep-Sea Research I* 53: 1344–1362.
- Beauchard, O., M. S. Thompson, K. E. Ellingsen, G. Piet, P. Laffargue & K. Soetaert, 2023. Assessing sea floor functional biodiversity and vulnerability. *Marine Ecology Progress Series* 708: 21–43.
- Beesley, P. L., G. J. Ross & C. J. Glasby, 2000. Polychaetes & Allies The Southern Synthesis. *Fauna of Australia: Polychaeta, Myzostomida, Pogonophora, Echiura, Sipuncula*, CSIRO Publishing, Australia: 420.
- Benavides-Serrato, M., G. Borrero-Pérez & C. M. Díaz-Sánchez, 2011. Equinodermos del Caribe colombiano I: Crinoidea, Asteroidea y Ophiuroida. *Instituto de Investigaciones Marinas y Costeras (Invemar) Colombia*. pp. 384.
- Ben-Haddad, M., M. Rida, S. Hajji, G. E. De-la-torre, H. Abou, N. Rangel-Buitrago & A. Ait, 2022. The wedge clam *Donax trunculus* L., 1758 as a bioindicator of microplastic pollution. *Marine Pollution Bulletin* 178: 13607.
- Bharath, M., U. Natesan, R. Vaikunth, P. Kumar, R. Ruthra & S. Srinivasalu, 2021. Spatial distribution of microplastic concentration around landfill sites and its potential risk on groundwater. *Chemosphere* 277: 130263. <https://doi.org/10.1016/j.chemosphere.2021.130263>.
- Blaskovic, A., P. Fastelli, H. Cizmek, C. Guerranti & M. Renzi, 2017. Plastic litter in sediments from the Croatian marine protected area of the natural park of Telašćica bay (Adriatic Sea). *Marine Pollution Bulletin* 114: 583–586. <https://doi.org/10.1016/j.marpolbul.2016.09.018>.
- Blott, S. J. & K. Pye, 2001. Gradistat: a grain size distribution and statistics package for the analysis of unconsolidated sediments. *Earth Surface Processes and Landforms* 26: 1237–1248.
- Botterell, Z. L., N. Beaumont, T. Dorrington, M. Steinke, R. C. Thompson & P. K. Lindeque, 2019. Bioavailability and effects of microplastics on marine zooplankton: a review. *Environmental Pollution* 245: 98–110. <https://doi.org/10.1016/j.envpol.2018.10.065>.
- Browne, M. A., P. Crump, S. J. Niven, E. Teuten, A. Tonkin, T. Galloway & R. C. Thompson, 2011. Accumulation of

- microplastic on shorelines worldwide: Sources and sinks. *Environmental Science & Technology* 45: 9175–9179.
- Brusca, R. C. & E. W. Iverson, 1985. A guide to the marine isopod Crustacea of Pacific Costa Rica. *Revista De Biología Tropical* 33: 1–77.
- Campos, N. H., G. R. Navas, A. Bermúdez & N. Cruz, 2005. Los Crustáceos Decápodos de la Franja Superior del Talud Continental (200–500 m) del Mar Caribe Colombiano, Universidad Nacional de Colombia, Facultad de Ciencias, Instituto de Ciencias Naturales, Bogotá., 272.
- Cantera, J. R. & J. F. Blanco, 2001. The estuary ecosystem of Buenaventura Bay, Colombia, Coastal Marine Ecosystems of Latin America Springer, Berlin: 265–280.
- Carreras-Colom, E., M. Constenla, A. Soler-Membrives, J. E. Cartes, M. Baeza, F. Padrós & M. Carrassón, 2018. Spatial occurrence and effects of microplastic ingestion on the deep-water shrimp *Aristeus antennatus*. *Marine Pollution Bulletin* 133: 44–52. <https://doi.org/10.1016/j.marpolbul.2018.05.012>.
- Castillo-Torres, P. A, 2002. Pastos marinos del Caribe colombiano. MSc Thesis. Universidad de Bogotá Jorge Tadeo Lozano.
- Choy, C. A. & J. C. Drazen, 2013. Plastic for dinner? Observations of frequent debris ingestion by pelagic predatory fishes from the central North Pacific. *Marine Ecology. Progress Series* 485: 155–163. <https://doi.org/10.3354/meps10342>.
- Compa, M., A. Ventero, M. Iglesias & S. Deudero, 2018. Ingestion of microplastics and natural fibres in *Sardina pilchardus* (Walbaum, 1792) and *Engraulis encrasicolus* (Linnaeus, 1758) along the Spanish Mediterranean coast. *Marine Pollution Bulletin* 128: 89–96. <https://doi.org/10.1016/j.marpolbul.2018.01.009>.
- Corti, A., J. La Nasa, G. Biale, A. Ceccarini, A. Manariti, F. Modugno & V. Castelvetro, 2023. Contaminación por microplásticos en los sedimentos de ambientes acuáticos interconectados de lechos lacustres, marinos y costeros: masa total específica de polímeros mediante el procedimiento multianalítico «PISA». *Química Analítica y Bioanalítica* 415: 2921–2936. <https://doi.org/10.1007/s00216-023-04664-0>.
- Costa, L. L., M. F. da Costa & I. R. Zalmon, 2021. Macroinvertebrates as biomonitors of pollutants on natural sandy beaches: overview and meta-analysis. *Environmental Pollution* 275: 116629. <https://doi.org/10.1016/j.envpol.2021.116629>.
- Choong, W. S., T. Hadi Barata, A. Yuniarto, K. H. Tang, F. Abdullah, M. Syafrudin, D. A. Al Farraj & A. M. Al-Mohaimeed, 2021. Characterization of microplastics in the water and sediment of Baram River estuary. *Borneo Island. Marine pollution bulletin* 172: 2880.
- Daccarett, E. Y. & V. M. Bossio, 2011. Colombian seashells from the Caribbean Sea. *L'informatore Piceno, Italia*. pp 384.
- Davinson, K. & S. E. Dudas, 2016. Microplastic Ingestion by Wild and Cultured Manila Clams (*Venerupis philippinarum*) from Baynes Sound, British Columbia. *Archives of Environmental Contamination and Toxicology* 71: 147–156.
- De León González, J. Á., J. R. Bastidas-Zavala, L. F. Carrera-Parra, M. E. García-Garza, V. Solís-Weiss & M. A. Tovar-Hernández, 2021. Anélidos Marinos de México y América tropical, Editorial Universitaria Universidad Autónoma de Nuevo León, México., 1049.
- Demographia, 2023. Demographia World Urban Areas: Built up Urban Areas Or World Agglomerations (19th ed.). Retrieved from <http://www.demographia.com/db-world-ua.pdf>.
- Departamento Administrativo Nacional de Estadística (DANE), 2025. Cuántos habitantes tendrá Buenaventura, Valle del Cauca en 2025: Proyecciones del Censo Nacional de Población y Vivienda del DANE Retrieved from <https://telencuestas.com/censos-de-poblacion/colombia/2025/valle-del-cauca/buenaventura>
- Desforges, J. P., M. Galbraith & P. S. Ross, 2015. Ingestion of microplastics by zooplankton in the Northeast Pacific Ocean. *Archives of Environmental Contamination and Toxicology* 69: 320–330.
- Díaz, J. M. & M. Puyana, 1994. Moluscos del Caribe colombiano. Un catálogo ilustrado, Editorial Presencia, Colombia., 291.
- Do, T. O., T. T. Duong, T. H. Nguyen, T. H. Hoang, D. H. Phung, M. V. Dang, M. T. N. Vu, T. P. Q. Le, V. C. Bui & H. Bui, 2022. Preliminary Investigation of Microplastics in Sediments from Industrial Manufacturing Waste Sources. *VNU Journal of Science: Natural Sciences and Technology* 38: 63–70. <https://doi.org/10.25073/2588-1140/vnunst.5336>.
- Duane, D. B., 1964. Significance of skewness in recent sediments, western Pamlico Sound, North Carolina. *Journal of Sedimentary Research* 34: 864–874.
- Enders, K., A. Käppler, O. Biniash, P. Feldens, N. Stollberg, X. Lange, D. Fisher, K. J. Eichhorn, F. Pollehne, S. Oberbeckmann & M. Labrenz, 2019. Tracing microplastics in aquatic environments base on sediment analogies. *Scientific Reports* 9: 1–15. <https://doi.org/10.1038/s41598-019-50508-2>.
- Esmeralda, G. & J. Patterson, 2025. Accumulation of microplastics in relation to the sediment grain size distribution in seagrass meadows in the coastal areas of Tuticorin, southeast coast of India. *Marine Pollution Bulletin* 212: 117557. <https://doi.org/10.1016/j.marpolbul.2025.117557>.
- Franco-Herrera, A., 2005. Oceanografía de la Ensenada de Gaira: El Rodadero, más que un centro turístico en el Caribe colombiano, Fundación Universidad de Bogotá Jorge Tadeo Lozano, Colombia., 56.
- Galindo-Montero, A. A., L. C. Costa-Redondo, O. Vasco-Echeverri & V. A. Arana, 2023. Microplastic pollution in coastal areas of Colombia: Review. *Marine Environmental Research* 190: 106027. <https://doi.org/10.1016/j.marenvres.2023.106027>.
- Gallitelli, L., A. Cera, G. Cesarini, L. Pietrelli & M. Scalici, 2021. Preliminary indoor evidences of microplastic effects on freshwater benthic macroinvertebrates. *Scientific Reports* 11: 720. <https://doi.org/10.1038/s41598-020-80606-5>.
- Galloway, T. S., M. Cole & C. N. Lewis, 2017. Interactions of microplastic debris throughout the marine ecosystem. *Nature Ecology & Evolution* 1: 1–8.
- Gamboa-García, D. E., G. Duque & P. Cogua, 2018. Dinámica de la estructura y composición de

- macroinvertebrados y su relación con las variables ambientales en la bahía de Buenaventura. Boletín de Investigaciones Marinas y Costeras. <https://doi.org/10.25268/BIMC.INVEMAR.2018.47.1.738>.
- Gamboa-García, D. E., G. Duque, P. Cogua & J. L. Marugo-Negrete, 2020. Mercury dynamics in macroinvertebrates in relation to environmental factors in a highly impacted tropical estuary: Buenaventura Bay, Colombian Pacific. *Environmental Science and Pollution Research* 27: 4044–4057. <https://doi.org/10.1007/s11356-019-06970-6>.
- García, E., A. E. Sandoval & G. Duque, 2021. Hábitos tróficos del lenguado *Citharichthys gilberti* (Paralichthyidae) y su relación con las dinámica ambiental en la Bahía de Buenaventura, Pacífico colombiano. *Boletín De Investigaciones Marinas y Costeras* 50: 133–150. <https://doi.org/10.25268/bimc.invemar.2021.50.2.1078>.
- García-Suárez, S. D., A. Acosta, E. Londoño-Cruz & J. Cantera, 2012. Organismos fósiles y móviles del litoral rocoso en el Pacífico Colombiano: Una guía visual para su identificación. *Invemar*, Colombia. pp.133.
- Gorska, B., S. Gromisz & M. Włodarska-Kowalczyk, 2019. Size assessment in polychaete worms—application of morphometric correlations for common North Atlantic taxa. *Limnology and Oceanography: Methods* 17: 254–265. <https://doi.org/10.1002/lom3.10310>.
- Green, B. C. & C. L. Johnson, 2020. Characterization of microplastic contamination in sediment of England inshore's waters. *Marine Pollution Bulletin* 151: 110788. <https://doi.org/10.1016/j.marpolbul.2019.110788>.
- Gurjar, U. R., K. A. Xavier, S. P. Shukla, A. K. Jaiswar, G. Deshmukhe & B. B. Nayak, 2022. Microplastic pollution in coastal ecosystem off Mumbai coast. *India. Chemosphere* 288: 132484. <https://doi.org/10.1016/j.chemosphere.2021.132484>.
- Hashim, S. M., M. J. Alam, Z. Yang & H. Arakawa, 2024. Microplastic assessment in the benthic ecosystem of Tokyo Bay: Sediment, water, and macrobenthic perspectives. *Regional Studies in Marine Science* 70: 103384. <https://doi.org/10.1016/j.rsma.2024.103384>.
- Hickman, C. P., 1998. *A Field Guide to Sea Stars and other Echinoderms of Galápagos*, Sugar Spring Press. United States of America, Galápagos Marine Life Series., 94.
- Hidalgo-Ruz, V., L. Gutow, R. C. Thompson & M. Thiel, 2012. Microplastics in the marine environment: a review of the methods used for identification and quantification. *Environmental Science & Technology* 46: 3060–3075. <https://doi.org/10.1021/es2031505>.
- Hiki, K. & F. Nakajima, 2022. Microplastic Ingestion by a Benthic Amphipod in Different Feeding Modes. *The Journal of Water and Environment Technology* 20: 37–144.
- Hodgson, D. 2018. The impacts of Microplastics Ingestion on Marine Polychaete Worms. MSc Thesis, University of Exeter.
- Holme, N. A. & A. D. McIntyre, 1984. *Methods for the study of marine benthos*, 2nd ed. Blackwell Scientific Publications, England.: 387.
- Horton, A. A., C. Svendsen, R. J. Williams, D. J. Spurgeon & E. Lahive, 2017. Large microplastic particles in sediments of tributaries of the river Thames, UK—abundance, sources and methods for effective quantification. *Marine Pollution Bulletin* 114: 218–226. <https://doi.org/10.1016/j.marpolbul.2016.09.004>.
- Hurley, R. R., J. C. Woodward & J. J. Rothwell, 2017. Ingestion of microplastics by freshwater Tubifex worms. *Environmental Science & Technology* 51: 12844–12851.
- Invemar, 2008. Evaluación preliminar de las amenazas naturales en la cuenca del río Manzanares con énfasis en eventos fluvio-torrenciales. *Ingeominas*, Colombia. pp. 79.
- Isobe, A., K. Uchida, T. Tokai & S. Iwasaki, 2015. East Asian seas: a hot spot of pelagic microplastics. *Marine Pollution Bulletin* 101: 618–623. <https://doi.org/10.1016/j.marpolbul.2015.10.042>.
- Isobe, A., K. Uchiyama-Matsumoto, K. Uchida & T. Tokai, 2017. Microplastics in the Southern Ocean. *Marine Pollution Bulletin* 114: 623–626.
- Ita-Nagy, D., I. Vázquez-Rowe & R. Kahhat, 2022. Prevalence of microplastics in the ocean in Latin America and the Caribbean. *Journal of Hazardous Materials Advances* 5: 100037. <https://doi.org/10.1016/j.hazadv.2021.100037>.
- Keen, A. M., 1971. *Sea Shells of Tropical West America: Marine Mollusks from Baja California to Peru*, 2nd ed. Stanford University Press, Stanford: 1064.
- Kensley, B. & M. Schotte, 1989. *Marine Isopod Crustaceans of the Caribbean*, Smithsonian Institution Press, Washington, DC., 293.
- Keszy, K., S. Oberbeckmann, B. Kreikemeyer & M. Labrenz, 2019. Spatial environmental heterogeneity determines young biofilm assemblages on microplastics in Baltic Sea mesocosms. *Frontiers in Microbiology* 10: 1665. <https://doi.org/10.3389/fmicb.2019.01665>.
- Knutsen, H. J. B., C. Cyvin, Ø. Totland, E. J. Lilleeng, V. Wade, A. Castro, J. Laugesen, Pettersen, T. Moskeland & H. P. Arp, 2020. Microplastic accumulation by tube-dwelling, suspension feeding polychaetes from the sediment surface: A case study from the Norwegian Continental Shelf. *Marine Environmental Research* 161: 105073. <https://doi.org/10.1016/j.marenvres.2020.105073>.
- Laverde-Castillo, J. J., 1986. Lista anotada de los poliquetos (Annelida) registrados para el Pacífico Colombiano, con notas preliminares sobre su zoogeografía. *Actualidades Biológicas* 15: 123–130. <https://doi.org/10.17533/udea.acbi.330103>.
- Lazarus, J. F. & J. R. Cantera, 2007. *Crustáceos (Crustacea: Sessilia, Stomatopoda, Isopoda, Amphipoda, Decapoda) de Bahía Málaga, Valle del Cauca (Pacífico colombiano)*. *Biota Colombiana* 8: 221–190.
- LeCroy, S. E., J. S. Richardson & D. Cobb, 2000. An illustrated identification guide to the nearshore marine and estuarine gammaridean Amphipoda of Florida Department of Environmental Protection, Division of Resource Assessment and Management. United States of America. pp. 2000.
- Lopes, L., I. D. da Costa, A. da Silva & I. Rosental, 2023. “Microplastic ecology”: Testing the influence of ecological traits and urbanization in microplastic ingestion by sandy beach fauna. *Estuarine, Coastal and Shelf Science* 290: 108406. <https://doi.org/10.1016/j.ecss.2023.108406>.
- Lopez, G. R. & J. S. Levinton, 2011. Particulate organic detritus and detritus feeders in coastal food webs. *The Treatise*

- on Estuarine and Coastal Science 6: 5–21. <https://doi.org/10.1016/B978-0-12-374711-2.00602-1>.
- Martins, J. & P. Sobral, 2011. Plastic marine debris on the Portuguese coastline: a matter of size? *Marine Pollution Bulletin* 62: 2649–2653.
- Mason, W. T. & P. P. Yevich, 1967. The use of Phloxine B and Rose Bengal stains to facilitate Sorting Benthic Samples. *Transactions of the American Microscopical Society* 86: 221–223. <https://doi.org/10.2307/3224697>.
- Mehdinia, A., R. Dehbandi, A. Hamzehpour & R. Rahnama, 2020. Identification of microplastics in the sediments of southern coasts of the Caspian Sea, north of Iran. *Environmental Pollution* 258: 113738. <https://doi.org/10.1016/j.envpol.2019.113738>.
- Mendes, A. M., N. Golden, R. Bermejo & L. Morrison, 2021. Distribution and abundance of microplastics in coastal sediments depends on grain size and distance from sources. *Marine Pollution Bulletin* 172: 112802. <https://doi.org/10.1016/j.marpolbul.2021.112802>.
- Molina, A., G. Duque & P. Cogua, 2020. Influences of environmental conditions in the fish assemblage structure of a tropical estuary. *Marine Biodiversity* 50: 5.
- Narmatha-Sathish, M., K. Immaculate Jeyasanta & J. Patterson, 2020. Monitoring of microplastics in the clam *Donax cuneatus* and its habitat in Tuticorin coast of Gulf of Mannar (GoM), India. *Environmental Pollution* 266: 115219. <https://doi.org/10.1016/j.envpol.2020.115219>.
- Pagter, E., R. Nash, J. Frias & F. Kavanagh, 2021. Assessing microplastic distribution within infaunal benthic communities in a coastal embayment. *Science of the Total Environment* 791: 148278. <https://doi.org/10.1016/j.scitotenv.2021.148278>.
- Paredes, C., J. Iannaccone & L. Alvarino, 2005. Use of benthic macroinvertebrates as bioindicators of water quality in Rímac River, Lima-Callao, Perú. *Revista Colombiana de Entomología* 31: 219–225.
- Pasquier, G., P. Doyen, N. Carlesi & R. Amara, 2022. An innovative approach for microplastic sampling in all surface water bodies using an aquatic drone. *Heliyon* 8: e11662. <https://doi.org/10.1016/j.heliyon.2022.e11662>.
- Peng, X., M. Chen, S. Chen, S. Dasgupta, H. Xu, K. Ta, M. Du, J. Li, Z. Guo & S. Bai, 2018. Microplastics contaminate the deepest part of the world's ocean. *Geochemical Perspectives Letters* 9: 1–5. <https://doi.org/10.7185/GEOCH-EMLET.1829>.
- Quirós-Rodríguez, J. A., C. Nisperuza-Pérez & J. Yepes-Esco-bar, 2021. Los microplásticos, una amenaza desconocida para los ecosistemas marinos de Colombia: perspectivas y desafíos a enfrentar. *Gestión y Ambiente* 24: 91615. <https://doi.org/10.15446/GA.V24N1.91615>.
- Rangel-Buitrago, N., H. Arroyo-Olarte, J. Trilleras, V. A. Arana, E. Mantilla-Barbosa, A. Gracia & A. Micallef, 2021. Microplastics pollution on Colombian Central Caribbean beaches. *Marine Pollution Bulletin* 170: 112685. <https://doi.org/10.1016/j.marpolbul.2020.111837>.
- Renzi, M., A. Blaskovic, P. Fastelli, M. Marcelli, C. Guer-ranti, S. Cannas, L. Barone & F. Massara, 2018. Is the microplastic selective according to the habitat? Records in *amphioxus* sands, Mäerl bed habitats and *Cymodocea nodosa* habitats. *Marine Pollution Bulletin* 130: 179–183. <https://doi.org/10.1016/j.marpolbul.2018.03.019>.
- Restrepo, J. D. & S. A. López, 2008. Morphodynamics of the Pacific and Caribbean deltas of Colombia, South America. *Journal of South American Earth Sciences* 25: 1–21.
- Ricaurte-Villota, C. & M.L. Bastidas-Salamanca, 2017. Regionalización oceanográfica: una visión dinámica del Caribe. Instituto de Investigaciones Marinas y Cos-teras José Benito Vives De Andr  is (INVEMAR). Santa Marta, Colombia. pp. 180.
- Rinc  n, C. H., G. Bol  var, R. Neira & E. J. Pe  a, 2008. Utilizaci  n de la macrofauna bent  nica como indicador de calidad ambiental en la desembocadura del R  o Anchicay  , Pac  fico colombiano. *Ingenier  a de Recursos Naturales y del Ambiente* 7: 94–101.
- Rus  nque-Quintero, L. L., G. A. Montoya-Rojas & A. L. Moyano-Molano, 2022. Environmental risks due to the presence of microplastics in coastal and marine environments of the Colombian Caribbean. *Marine Pollution Bulletin* 185: 114357. <https://doi.org/10.1016/j.marpolbul.2022.114357>.
- Sahu, B. K., 1964. Depositional mechanisms from the size analysis of clastic sediments. *Journal of Sedimentary Research* 34: 73–83.
- Set  l  , O., J. Norkko & M. Lehtiniemi, 2016. Feeding type affects microplastic ingestion in a coastal invertebrate community. *Marine Pollution Bulletin* 102: 95–101. <https://doi.org/10.1016/j.marpolbul.2015.11.053>.
- Sly, P. G., R. L. Thomas & B. R. Pelletier, 1982. Comparison of sediment energy-texture relationships in marine and lacustrine environments. *Hydrobiologia* 91: 71–84. <https://doi.org/10.1007/BF02391923>.
- Song, Y. K., S. H. Hong, M. Jang, J. H. Kang, O. Y. Kwon, G. M. Han & W. J. Shim, 2014. Large accumulation of micro-sized synthetic polymer particles in the sea surface microlayer. *Environmental Science & Technology* 48: 9014–9021.
- Thomas, Y. F., C. Ricaurte-Villota & O. Coca-Dom  nguez, 2014. Atlas climatol  gico del oc  ano Pac  fico colombiano y   reas adyacentes. Serie de Publicaciones Especiales No. 31. Santa Marta. pp. 170.
- Thompson, R. C., Y. Olsen, R. P. Mitchell, A. Davis, S. J. Rowland, A. W. John, D. McGonigle & A. E. Russell, 2004. Lost at sea: Where is all the plastic? *Science* 304: 838–838.
- V  squez-Molano, D., A. Molina & G. Duque, 2021. Distribuci  n espacial y aumento a trav  s del tiempo de micropl  sticos en sedimentos de la Bah  a de Buenaventura, Pac  fico colombiano. *Bolet  n De Investigaciones Marinas y Costeras* 50: 27–42. <https://doi.org/10.25268/BIMC.INVEMAR.2021.50.1.1021>.
- Vecchi, S., J. Bianchi, M. Scalici, F. Fabroni & P. Tomas-setti, 2021. Field evidence for microplastic interactions in marine benthic invertebrates. *Scientific Reports* 11: 20900. <https://doi.org/10.1038/S41598-021-00292-9>.
- Vianello, A., A. Boldrin, P. Guerriero, V. Moschino, R. Rella, A. Sturaro & L. Da Ros, 2013. Microplastic particles in sediments of Lagoon of Venice, Italy: First observations on occurrence, spatial patterns, and identification. *Estuarine, Coastal and Shelf Science* 130: 54–61. <https://doi.org/10.1016/j.ecss.2013.03.022>.
- Vidal, L., A. E. M. Sandoval & G. Duque, 2021. Incremento de la contaminaci  n por micropl  sticos en aguas

- superficiales de la bahía de Buenaventura, Pacífico colombiano. *Boletín de Investigaciones Marinas y Costeras* 50: 113–132.
- Visser, G. S., 1969. Grain size distributions and depositional processes. *Journal of Sedimentary Research* 39: 1074–1106.
- Wetzer, R., R. Brusca & G. Wilson, 1997. Taxonomic Atlas of the Benthic Fauna of the Santa Maria Basin and Western Santa Barbara Channel. Vol. 11 - The Crustacea, Part 2. The Isopoda, Cumacea, and Tanaidacea. Santa Barbara Museum of Natural History. Santa Barbara, United States of America. pp 278.
- Wicaksono, E. A., S. Werorilangi, T. S. Galloway & A. Tahir, 2021. Distribution and seasonal variation of microplastics in Tallo River, Makassar. Eastern Indonesia. *Toxics* 9: 129. <https://doi.org/10.3390/toxics9060129>.
- Woodall, L. C., A. Sanchez-Vidal, M. Canals, G. L. Paterson, R. Coppock, V. Sleight, A. Calafat, A. D. Rogers, B. E. Narayanaswamy & R. Thompson, 2014. The deep sea is a major sink for microplastic debris. *Royal Society Open Science* 1: 140317. <https://doi.org/10.1098/rsos.140317>.
- Zhang, Q., J. Xie, S. Ma, Y. Chen, F. Lin & X. Diao, 2023. Occurrence and characteristics of microplastics in benthic species from mangrove wetlands of Hainan. South China. *Frontiers in Marine Science* 10: 965059.
- Zhang, W., S. Zhang, J. Wang, Y. Wang, J. Mu, P. Wang & D. Ma, 2017. Microplastic pollution in the surface waters of the Bohai Sea, China. *Environmental Pollution* 231: 541–554. <https://doi.org/10.1016/j.envpol.2017.08.058>.
- Zhao, J., W. Ran, J. Teng, Y. Liu, H. Liu, X. Yin & Q. Wang, 2018. Microplastic pollution in sediments from the Bohai Sea and the Yellow Sea, China. *Science of the Total Environment* 640: 637–645. <https://doi.org/10.1016/j.scitotenv.2018.05.346>.
- Zheng, T., 2022. Global sources, abundance, size, and distribution of microplastics in marine sediments – a critical review. *Estuarine, Coastal and Shelf Science* 264: 107702. <https://doi.org/10.1016/j.ecss.2021.107702>.

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